The Sequential Operator on Containers: Units, Associator, and Composition of Extensions

Extracted from a machine-checked development

Abstract

We equip the category of containers with the *sequential operator* $\triangleleft$, whose extension is composition of the induced polynomial functors: $\mathrm{Ext}(G \triangleleft F) \cong \mathrm{Ext}\, G \circ \mathrm{Ext}\, F$. The unit is the identity container $I = 1 \triangleleft !$. We construct the left and right unitors $I \triangleleft F \cong F$ and $F \triangleleft I \cong F$, and the associator $(H \triangleleft G) \triangleleft F \cong H \triangleleft (G \triangleleft F)$, as isomorphisms of containers. All dependent types involved agree definitionally: the associator is re-association of an iterated dependent sum, and no transport is required. A $\triangleleft$-comonoid in the category of containers is precisely a directed container. All stated results have been machine-checked.

Contents

1 Introduction **1**

2 Containers recalled **2**

3 The sequential operator **2**

4 Extension is functor composition **2**

5 Isomorphisms of containers **3**

6 Left and right unitors **3**

7 The associator **4**

8 Organising remark: comonoids and directed containers **4**

9 Discussion **4**

1 Introduction

Containers present a class of polynomial endofunctors on types: a shape type equipped with a family of position types determines the functor

$$X \longmapsto \sum_{s} X^{P(s)}.$$

Composition of such functors is again a polynomial functor of the same form, and the composite has an explicit combinatorial description. That description is the sequential operator $\triangleleft$.

The operator is the spine of the theory of directed containers: a directed container is exactly a $\triangleleft$-comonoid in the category of containers. The present note records the monoidal structure itself—the operator, its unit, and the coherence isomorphisms—leaving the comonoid laws to the directed-container development.

What this note claims. In ordinary mathematical language: the sequential composite $G \triangleleft F$ has shapes

$$\sum_{t\,:\,\mathrm{Shape}(G)} \big(\mathrm{Pos}(G)(t) \to \mathrm{Shape}(F)\big)$$

and positions that are pairs of an outer G-position with an inner F-position in the shape sitting there. Its extension is functor composition. The monoidal coherences (unitors and associator) are isomorphisms of containers whose underlying shape maps are the evident bijections, and all position fibres line up definitionally.

2 Containers recalled

Definition 2.1 (Container)**.** A **container** C consists of a type $\mathrm{Shape}(C)$ of shapes and, for each shape s, a type $\mathrm{Pos}(C)(s)$ of positions. We write $C = (S, P)$ when convenient.

Definition 2.2 (Extension)**.** The **extension** of a container C at a type X is

$$\mathrm{Ext}\, C\, X \;=\; \sum_{s\,:\,\mathrm{Shape}(C)} \big(\mathrm{Pos}(C)(s) \to X\big). \tag{1}$$

Definition 2.3 (Container morphism)**.** A **morphism** $\varphi : C \Rightarrow D$ consists of a map on shapes $u : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$ and, for each shape s, a contravariant map on positions $g_s : \mathrm{Pos}(D)(u(s)) \to \mathrm{Pos}(C)(s)$.

We write **Cont** for the category of containers and container morphisms. Composition of morphisms and identities are the evident ones (covariant on shapes, contravariant on positions).

3 The sequential operator

Definition 3.1 (Sequential operator)**.** Let F and G be containers. The **sequential composite** $G \triangleleft F$ ("G outside, F inside") is the container with

$$\mathrm{Shape}(G \triangleleft F) \;=\; \sum_{t\,:\,\mathrm{Shape}(G)} \big(\mathrm{Pos}(G)(t) \to \mathrm{Shape}(F)\big), \tag{2}$$

$$\mathrm{Pos}(G \triangleleft F)\,(t, f) \;=\; \sum_{q\,:\,\mathrm{Pos}(G)(t)} \mathrm{Pos}(F)\big(f(q)\big). \tag{3}$$

A shape is an outer G-shape together with a choice of inner F-shape for each G-position; a position is an outer position together with an inner position in the shape sitting there.

Definition 3.2 (Identity container)**.** The **identity container** is

$$I \;=\; (1,\; * \mapsto 1), \tag{4}$$

one shape and one position. Its extension is the identity functor, and it is the unit for $\triangleleft$.

4 Extension is functor composition

The defining property of $\triangleleft$ is that extension turns sequential composition into composition of functors.

Definition 4.1 (Uncurrying of labellings)**.** For containers G and F and a type X, define maps

$$\alpha_X \;:\; \mathrm{Ext}\,(G \triangleleft F)\, X \;\longrightarrow\; \mathrm{Ext}\, G\,(\mathrm{Ext}\, F\, X), \tag{5}$$

$$\beta_X \;:\; \mathrm{Ext}\, G\,(\mathrm{Ext}\, F\, X) \;\longrightarrow\; \mathrm{Ext}\,(G \triangleleft F)\, X \tag{6}$$

by regrouping: α sends a labelling of the nested position sum to an outer structure of inner structures; β is the inverse uncurrying. Explicitly, writing a shape of $G \triangleleft F$ as $\langle t, f\rangle$ and a labelling as v,

$$\alpha_X \langle \langle t, f\rangle, v\rangle \;=\; \langle t,\; q \mapsto \langle f(q),\; r \mapsto v\langle q, r\rangle\rangle\rangle, \tag{7}$$

$$\beta_X \langle t, m\rangle \;=\; \langle \langle t,\; q \mapsto \pi_1(m(q))\rangle,\; \langle q, r\rangle \mapsto \pi_2(m(q))(r)\rangle. \tag{8}$$

Theorem 4.2 (Extension of sequential composite)**.** *For all containers G and F and every type X, the maps α_X and β_X are mutually inverse:*

$$\beta_X \circ \alpha_X \;=\; \mathrm{id}, \qquad \alpha_X \circ \beta_X \;=\; \mathrm{id}. \tag{9}$$

Thus $\mathrm{Ext}\,(G \triangleleft F) \cong \mathrm{Ext}\,G \circ \mathrm{Ext}\,F$ *as functors on types. Both fibres agree definitionally, so the round trips hold after case analysis on nested pairs.*

Proof. By lean. □

5 Isomorphisms of containers

Definition 5.1 (Container isomorphism)**.** An **isomorphism of containers** $C \cong D$ is a pair of morphisms $\mathrm{hom} : C \Rightarrow D$ and $\mathrm{inv} : D \Rightarrow C$ that are mutual inverses under composition of container morphisms.

Remark 5.2. The unitors and associator are isomorphisms, not equalities: the underlying shape types are bijective (e.g. $(1 \to S) \cong S$) but not definitionally identical.

6 Left and right unitors

Theorem 6.1 (Left unitor)**.** *For every container F there is an isomorphism of containers*

$$I \triangleleft F \;\cong\; F. \tag{10}$$

Forward, read off the single inner shape ($f : 1 \to \mathrm{Shape}(F)$ maps to $f(*)$); backward, send a shape s to the constant assignment $* \mapsto s$. Positions transport along the same collapse of the unit type. Both round trips hold by η-laws for the unit type and dependent sums.

Proof. By lean. □

Theorem 6.2 (Right unitor)**.** *For every container F there is an isomorphism of containers*

$$F \triangleleft I \;\cong\; F. \tag{11}$$

Forward, drop the trivial inner data ($g : \mathrm{Pos}(F)(s) \to 1$ is unique); backward, pad with the unique map to the unit type. Positions $\sum_{p\,:\,\mathrm{Pos}(F)(s)} 1$ collapse to $\mathrm{Pos}(F)(s)$.

Proof. By lean. □

7 The associator

Theorem 7.1 (Associator)**.** *For all containers H, G, and F there is an isomorphism of containers*

$$(H \triangleleft G) \triangleleft F \;\cong\; H \triangleleft (G \triangleleft F). \tag{12}$$

Proof. By lean. □

Remark 7.2 (Explicit data)**.** A shape of the left-hand side is a triple $(\langle a, g\rangle, k)$ with $g : \mathrm{Pos}(H)(a) \to \mathrm{Shape}(G)$ and

$$k \;:\; \Big(\sum_{u \,:\, \mathrm{Pos}(H)(a)} \mathrm{Pos}(G)\big(g(u)\big) \Big) \to \mathrm{Shape}(F).$$

A shape of the right-hand side is a pair $\langle a, m\rangle$ with

$$m \;:\; \mathrm{Pos}(H)(a) \to \sum_{b \,:\, \mathrm{Shape}(G)} \big(\mathrm{Pos}(G)(b) \to \mathrm{Shape}(F)\big).$$

The bijection curries the inner data:

$$m(u) \;=\; \langle g(u),\; v \mapsto k\langle u, v\rangle\rangle, \tag{13}$$

inverted by $g(u) = \pi_1(m(u))$ and $k\langle u, v\rangle = \pi_2(m(u))(v)$.

On positions the isomorphism is re-association of a nested dependent sum,

$$\big\langle\langle u, v\rangle,\, p\big\rangle \;\longleftrightarrow\; \big\langle u,\, \langle v, p\rangle\big\rangle. \tag{14}$$

Crucially, the fibre $\mathrm{Pos}(F)\big(\pi_2(m(u))(v)\big)$ is definitionally equal to $\mathrm{Pos}(F)\big(k\langle u, v\rangle\big)$, so no transport is needed and the round trips close by case analysis on nested pairs.

8 Organising remark: comonoids and directed containers

Remark 8.1 ($\triangleleft$-comonoids)**.** A **$\triangleleft$-comonoid** in **Cont** is a container C equipped with a counit $C \Rightarrow I$ and a comultiplication $C \Rightarrow C \triangleleft C$ satisfying the comonoid laws (coassociativity up to the associator; counitality up to the unitors). Such a structure is precisely a **directed container**: equivalently, a comonad on types whose underlying functor is a container extension.

Thus $(\mathbf{Cont}, \triangleleft, I)$ is the monoidal home of directed containers. The comonoid laws unfold to the directed-container axioms (root, subtree, shift, and laws D1–D5); we record the correspondence as the organising remark and leave the equivalence to the directed-container development.

9 Discussion

What has been established. The sequential operator equips containers with a monoidal structure whose extension is functor composition. The unitors and associator are isomorphisms of containers; all relevant fibres agree definitionally, so the constructions are transport-free.

Why definitional agreement matters. In dependent type theory, re-associating nested sums often forces transport along equalities of fibres. Here the position type of $G \triangleleft F$ is arranged so that the two associations of a triple sum are definitionally the same family. The associator is therefore nothing but the combinatorial re-bracketing $((u, v), p) \leftrightarrow (u, (v, p))$.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows the theory of containers of Abbott, Altenkirch, and Ghani, and the sequential (composition) product standard in that literature. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

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